MEASURING THE IMPACT OF THE TOXICS RELEASE INVENTORY: EVIDENCE FROM MANUFACTURING PLANT BIRTHS

by

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Abstract

The Toxics Release Inventory was the first major initiative to take a disclosure-based approach to environmental regulation and has served as the model for several other disclosure-based environmental policies. Yet the magnitude of its direct impacts on industrial manufacturing outcomes has not been established. I use Census Bureau micro-data to estimate the impacts of the Toxics Release Inventory on the opening of new manufacturing plants. I find that on average, counties that were found to be among the dirtiest in the country, in terms of toxic emissions, experienced a decrease in "dirty" plant births and an even larger increase in "clean" plant births. Furthermore, the magnitude of this shift is closely related to per capita income in the affected coun- ties - the effect is strongest in high-income communities and is reversed in low-income communities. I discuss the implications for information-based environmental policies.[i]

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1 Introduction

Pollution disclosure has often been called the third wave[1] of environmental regulation. After earlier emphases on command and control regulation and market-based approaches, an increasing number of policies in an increasing number of jurisdictions are focusing on information disclosure as a primary tool. This approach has gained popularity partly because it does not force policy-makers to pick technology winners or impose uniform standards on firms and industries who face different costs of being "clean" (as can be the case in command and control regulation). Nor does it rely on a need to choose the "optimal" amount of pollution or optimal tax rates, as is required of the most common market-based solutions. Instead, it relies on a Coasian bargaining argument; if stakeholders and the general public value environmental attributes and have adequate information as to the polluting activities of firms, then the public can process this information via various channels, and impose costs or otherwise give incentives for firms to achieve the right balance between "clean" and "dirty". Incidentally, information provision is also generally less costly to administer than the methods employed in the two previous waves of regulation, another factor which undoubtedly plays a role in its proliferation as a policy tool.

The flagship example of a disclosure-based environmental policy is the Toxics Release Inventory (TRI), a program which has existed for more than 20 years. Created by the October 1986 passage of the Emergency Planning and Community Right-to-Know Act (EPCRA), the TRI is a publicly available database maintained by the U.S. Environmental Protection Agency (EPA) that contains information on toxic chemical releases and waste management activities reported annually by certain industries as well as federal facilities. Industrial or federal facilities that either produce more than 25,000 pounds or handle more than 10,000 pounds of any of the more than 600 listed toxic chemicals must provide detailed information

[1] Tietenberg (1998)

on the treatment, recycling, or release of these substances.[2] Every year, the TRI typically contains approximately 80,000 facility-chemical reports from more than 20,000 different facilities.

Since the first TRI data became publicly available in June of 1989, it has given rise to a vast body of anecdotal evidence suggesting that media, investors, workers, industry, the government, and the general public have used TRI data to learn about environmental risks and facility-level industrial performance. The EPA has documented more than 100 uses of the TRI data by government, businesses, and citizens (U.S. Environmental Protection Agency, 2003), and several of these documented uses have in turn led to increased efforts by companies to improve their environmental performance (Hamilton, 2005).

However, statistical evidence that the introduction of a disclosure-based approach led to systematic change in pollution outcomes is weak, due to the simple fact that the data covered in the TRI were not recorded in any widespread way prior to the implementation of the program. Rather, the existing research on the TRI (e.g., Hamilton 1995, Konar and Cohen 1996, Khanna et al. 1998, Bui and Mayer 2003, Oberholzer-Gee and Mitsunari 2006) has focused either on the reaction of different stakeholders to the release of information contained in the TRI, or the heterogeneity in improvements in environmental performance during the early years of the TRI. However, none have addressed the question of whether firms or plants were actually doing anything differently in response to the existence of the TRI. A primary contribution of this paper is that, in contrast with the studies listed above, I do not focus on only one channel of impact, as do the five studies above, nor am I limited in looking at only publicly traded firms. Rather, by focusing on manufacturing outcomes, I capture all channels, as well as the effects on all firms.

Traditional environmental regulation is thought to have a direct effect on manufacturing

[2]See Bennear (2008) for a thorough explanation of the factors that determine whether a facility is subject to the TRI disclosure requirements.

outcomes (in established industries) in three ways:

- it can force incumbents in the affected industries to close;
- it can induce incumbent plants in the affected industries to operate in a more environmentally-friendly way, either through changes in the production process or through improvements in pollution abatement; or
- it can prevent, or otherwise dissuade, would-be establishments from entry, especially in the dirtiest industries.

This paper investigates the third pathway to affect manufacturing, entry deterrence. Accordingly, my empirical analysis investigates whether we observe a shift in the patterns of new plant creation, or plant 'births' in the industrial sectors most likely to be affected following the implementation of the TRI. Previous literature (Becker and Henderson, 2000) has established a clear reduction in plant births in counties affected by traditional command-and-control regulation, but the question of whether a disclosure-based policy - which, unlike the Clean Air Act Amendments that Becker and Henderson (2000) analyze, does not allow the federal government to impose tough sanctions for non-attainment - has similar effects has not been answered. However, this benchmark for comparison is not the only advantage to focusing on plant births; while from an environmental standpoint one might think that the gains would be higher by closing an older, less efficient plant or changing the way these older plants operate, prevention of new plants from opening in the region may be more realistic. Incumbent plants often have well-entrenched political interests and also tend to be large employers within the community, both of which can render political action against them more costly than action against potential entrants.

I use unique establishment-level micro-data from the U.S. Census Bureau to address this question. I find that in the counties that had the highest levels of toxic releases, this distinction triggered a slight decrease in manufacturing plant births in high-TRI sectors, and

a simultaneous increase in the number of births in the cleanest industries. Furthermore, I find that these average effects mask a significant amount of heterogeneity in county-level 'responses' to the TRI, and I provide evidence that the magnitude of these shifts are highly correlated with the prevailing income levels in the county. These findings shed light not only on the question of disclosure-based environmental regulation, but also on the pollution haven hypothesis, which posits that equilibrium capital flows (represented here by new plant births) are affected by environmental regulations.

The remainder of this paper will be structured as follows. In section 2, I present a brief background on the relevant institutions of environmental regulation in the United States, while in section 3 I review the literature on the TRI and pollution disclosure programs more broadly. In section 4, I provide more detail on the Census Bureau data used here, while section 5 presents my empirical strategy to identify measurable effects of the TRI on plant births. Section 6 presents the results and Section 7 concludes the paper with a discussion of the implications of the findings herein.

2 Background

The institutions surrounding the regulation of pollution in the United States are complicated. In order to provide a better understanding of the context under which this disclosure-based approach to regulation was introduced, some aspects of the relevant institutional landscape are useful to highlight.

I begin by presenting a brief history of the environmental regulation of industrial pollution in the U.S. in the period leading up to the passage of EPCRA and the TRI. The Clean Air Act Amendments of 1977 were the fifth major piece of federal legislation relating to the reduction of air pollution since the original Air Pollution Control Act of 1955. The 1977

Amendments led to major changes in the way air pollution (though *not* toxic pollution[3]) was regulated in the United States. Under this regulatory regime (which continues, with minor modifications, today), counties whose ambient air quality is below federal standards for any of 6 "criteria" air pollutants can be designated as "non-attainment" counties. The states where these counties are located are required to develop State Implementation Plans that detail how these counties will attain federal air quality standards within 10 years. The Act granted the federal government the power to withhold federal highway funds or to impose moratoria on new plant construction in recalcitrant states. The bill also gave the federal government the power to impose civil penalties directly on polluters. The key regulatory instrument targeting industrial facilities was technological controls on equipment, under which both new and existing plants were subject to tighter standards in non-attainment areas. In attainment areas, only new large plants were subject to regulations.

The focus of EPCRA was very different. Motivation for increased federal regulation of toxic chemicals was triggered by a tragic chemical accident in Bhopal, India, in December 1984, and subsequent accidents in the U.S. that, while smaller in scale than Bhopal, nonetheless suggested vulnerability to similar accidents. EPCRA's purpose was to encourage and support emergency planning at state and local levels and to provide the public and government with information regarding possible risks from toxic emissions. The portion of the Act relevant to this study is Section 313, which requires industrial facilities that use, manufacture, or process above a threshold amount of any of the listed toxic chemicals in a given year to provide detailed information on the use and disposal of these chemicals to both the EPA and state officials by July 1 of the following year. This information is subsequently made available to the public, and is known as the Toxics Release Inventory. Data from the first reporting year, 1987, was released on June 19, 1989. This reporting lag has persisted

[3]Toxic, or hazardous, pollutants, as opposed to conventional pollutants, are those pollutants that are known or suspected to cause cancer or other serious health effects, such as reproductive effects or birth defects.

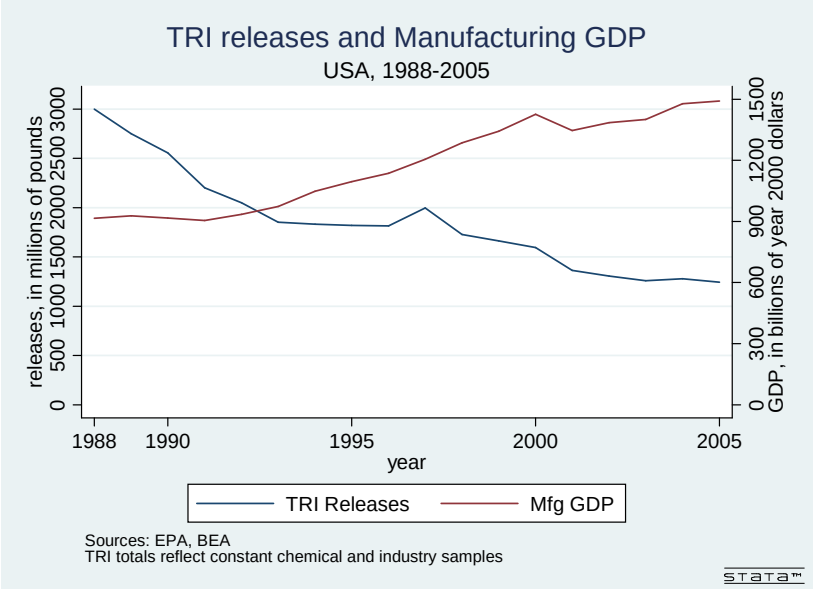


Figure 1: TRI Releases and Manufacturing GDP: USA, 1988-2005

throughout the period of the current study, with a given calendar year's data being released in the spring two years later.

The first two decades of the Toxics Release Inventory have coincided with a noticeable decrease in the amount of reported TRI releases; total releases have decreased by 58.5% between 1988 and 2005.[4] As Figure 4-1 indicates, this decrease has happened despite an increase in real manufacturing GDP during the same period.

A typical year's inventory contains approximately 80,000 facility-chemical reports from over 20,000 facilities. These data are often aggregated to identify the most highly-polluting facilities in various industries or geographic regions, or to benchmark corporate environmental performance, either within an industry or within a firm over time. The annual EPA

[4]While 1987 was the first year for which TRI data were collected, the data are generally regarded as problematic in some respects (U.S. General Accounting Office, 1991). As a result, analysis that examines decreases in TRI releases (e.g., Hamilton 1999) generally uses 1988 data as the baseline. Bennear (2008) points out that the method of aggregation used by EPA in calculating these figures assumes that a facility's failure to report a given chemical means that the facility's emissions of the chemical were 0, where in several cases failure to report will be positive, but under the threshold. She finds that up to 40% of the observed decline in reported toxic releases in Massachusetts may be attributed to non-reporting due to the reporting thresholds.

reports on TRI[5] also typically contain similar lists for the "dirtiest" counties. The existence of these lists suggest that facilities face varying degrees of public pressure to reduce toxic releases. These lists motivate the variation I exploit for identification in my empirical analysis.

The TRI is comprised of self-reported estimated emissions, and previous researchers have raised questions about its validity (see, e.g., Toffel and Marshall 2004). While the degree to which the TRI data are accurate remains an open question, there is a fair amount of quality control and verification before the data is released to the public, a fact which should allay some of these concerns.[6] EPA has also conducted TRI data quality surveys, most recently in 1996, which sent surveyors to several facilities to perform their own estimates of releases and other toxic waste management activities. These reports conclude that the information provided by respondents is generally fairly accurate. While facilities in the three sectors assessed in 1996 underestimated releases by 28% on average, three of the four previous surveys found that facility estimates were within 2.2%, on average, of the site surveyor estimates (and that over-reporting is nearly as frequent as under-reporting). These same surveys found that facilities correctly determined whether or not a chemical used at the

[5]See, for example, the 1987 TRI Public Data Release (EPA 1989). The TRI data were initially made available in paper reports, but eventually were also disseminated via CD-ROM's and the EPA website.

[6]I provide some more detail on the data collection and verification process here. Facilities must file annual reports on releases and waste treatment with EPA by July 1st of the following year. The U.S. EPA then checks the data for consistency, 'echoes' the information back to the facilities, allows facilities to make corrections or clarifications of errors, and will refer the issue to an EPA Regional Office for further investigation if necessary. For example, EPA's Region 5, which contains roughly one quarter of the facilities reporting to TRI, receives "several hundred" such referrals per year (Codina, 26 October 2009). Each EPA region also has a TRI enforcement program, which conducts a limited number of data quality inspections (of reporting facilities) and non-reporting inspections (of facilities that are in TRI industries but did not report) each year. Violations, whether stemming from late reporting, failure to report, or data quality issues, can lead to penalties of $25,000 per day, per chemical, per violation, and possibly criminal charges as well. The EPA routinely takes enforcement actions against facilities that either fail to report or report data inaccurately, suggesting that the incentive to provide truthful information is real. For example, in October 2009 EPA's Region 9 levied a $194,000 fine in October 2009 against a California facility that failed to report TRI chemicals over a five-year period (see U.S. Environmental Protection Agency 2009). In addition, EPCRA allows citizens to file suit for specified violations of the law, including reporting requirements (see Gray 2002, p.55-59).

facility was above the reporting threshold between 93 and 95% of the time (where, again, over-reporting is nearly as common as under-reporting).[7]

There are several anecdotal examples of the TRI data having been used by citizen groups, non-governmental organizations, or local regulators to spur some of the highest toxic emitters into reducing their toxic pollution, whether through citizen-led information campaigns, guides produced by national environmental organizations, or direct negotiation. After Calhoun County, Texas was highlighted by the initial TRI report as the county with the highest level of toxic releases in the country, a community group there used TRI data to build community awareness about local pollution, eventually obtaining a commitment from Alcoa for aggressive pollution reduction initiatives at two local plants. Similarly, a group in Butler County, PA, which was among the highest TRI counties for several years in the mid-to-late 1990's, eventually procured a commitment from the state to restrict the level of nitrates that a major steel plant in the area was allowed to release into Connoquenessing Creek. The section 313 reporting requirements have also been cited as directly influencing firms that produce and use TRI-designated chemicals, leading them to implement new emissions reduction initiatives (Baram et al. 1992, p.40; Hamilton 2005), sometimes even without external pressure (Hamilton 2005, p.4).

The existence of the TRI is also credited with leading to further changes in federal law on toxic pollution. The first, known as the Pollution Prevention Act of 1990, simply increased the reporting requirements of the TRI, mandating more information on the steps taken by the companies to reduce their generation of toxic materials. The second, passed in the same year is known as the Clean Air Act Amendments of 1990 (CAAA90). These Amendments included provisions related to the regulation of toxics, centered on requirements that the EPA develop maximum achievable control technology (MACT) standards that all new and

[7]It is also worth noting that despite any potential problems with TRI data, it has been widely used in both the economics and management literature as a measure of environmental performance (e.g., see Hamilton 1995, Konar and Cohen 1996, Khanna and Damon 1999, King and Lenox 2000, and Maxwell et al. 2000).

existing major source facilities would eventually need to meet. The Water Quality Act of 1987 also established stricter control of the release of toxic pollutants into bodies of water. None of the subsequent federal regulations appear to provide for any geographic heterogeneity in the regulatory pressures faced by either new or incumbent manufacturing plants.

3 Literature Review

There is a growing literature on the effects of TRI and similar pollution disclosure programs. I highlight those studies, as well as relevant studies from other strands of the environmental economics literature, in this section.

The theoretical basis for this type of program is derived from Coase (1960), who argues that with well-defined property rights, coupled with trade-ability of the externality and no transaction costs, efficient pollution levels should be reached. Resolution of the information asymmetry between polluter and stakeholder (whether that stakeholder is represented by the government or the affected public) should significantly lower one of the key transaction costs, implying that in certain cases a disclosure program may result in pollution levels that are closer to the efficient level. Accordingly, if the information asymmetry plays an important role in determining pollution outcomes, we should see some shift in these outcomes once the information asymmetry is reduced. Tietenberg (1998) identifies channels through which public disclosure may motivate improved environmental performance. These include output market pressures, input market pressures, judicial pressures, and regulatory pressures.

The empirical literature on the effects of the Toxic Release Inventory is sizeable, although the focus has not been on manufacturing outcomes *per se*. Hamilton (1995), Konar and Cohen (1996), and Khanna et al. (1998) find some evidence of a negative financial market response to the pollution 'news' of the early years of the TRI, and the latter two also find that these equity market reactions in turn had an effect on firm environmental performance in

subsequent years. Bui and Mayer (2003) and Oberholzer-Gee and Mitsunari (2006) instead focus on the effects of increased pollution transparency on housing prices; the former finds no effects while the latter identifies a significant response. Finally, Decker (2003), among other results, finds evidence that the time until permit approval for new plants under certain environmental statutes is a function of TRI releases in other facilities owned by that applying firm.[8]

Despite the fact that the effects of the TRI on firm behavior and pollution outcomes are not fully understood, disclosure programs have continued to proliferate as alternatives to traditional environmental regulation, particularly in the developing world. Blackman et al. (2004) surveys managers of plants participating in an Indonesian disclosure and rating program; their findings suggest that an important means by which the program spurs abatement is through improving managerial information. García et al. (2009) identify characteristics of Indonesian plants that were more responsive to these same ratings and find that foreign-owned plants, those in more densely populated areas, and those with low initial ratings were more responsive, all other things equal. Powers et al. (2008) analyzes a similar ratings program focusing on pulp and paper mills in India and finds that stand-alone plants, those in wealthier communities, and those with low initial ratings were more responsive.

More broadly, several studies have found strong relationships between environmental outcomes and community characteristics. In the absence of environmental regulation, this is often referred to as the 'informal regulation' hypothesis, and studies have found that environmental risks are decreasing in community income levels (Pargal and Wheeler 1996, Arora and Cason 1999) or propensity for collective action (Hamilton 1993). This informal regulation hypothesis has been extended in attempts to explain heterogeneity in the plant-level response to environmental policies and other policies that result in environmental outcomes, through

[8]This is consistent with anecdotal evidence, provided in U.S. Environmental Protection Agency (2003), of more than 40 uses of TRI data by state and federal regulators and legislators for the purpose of enforcement targeting, environmental risk assessment, crafting new legislation, and other uses.

the inclusion of interaction variables between the policy variable and variables that reflect community or plant/firm characteristics (Powers et al. 2008, García et al. 2009, Delmas et al. 2007, Delmas et al. 2010).

Establishment-level micro-data from the Census Bureau has previously been used to measure the effects of more conventional regulation. In particular, a series of papers has exploited the county-level variation in regulatory status under the 1977 Clean Air Act Amendments to identify its effects on county-level manufacturing outcomes. Becker and Henderson (2000) document a shift away from plant births in non-attainment counties for industries likely to be affected by the non-attainment designation, documenting reductions in plant birth rates of 26-45% in polluting industries. They also find evidence that regulatory status affected the size of new plants and the timing of investment. Greenstone (2002) uses Census of Manufactures data to estimate the effects of regulatory status on employment growth, capital formation, and output growth in pollution-intensive industries, and finds sizeable negative (though not always significant) effects on each. List et al. (2003) reinforce the findings in Becker and Henderson (2000), using a New York State dataset and semi-parametric matching methods. Becker (2005) shows that heavy emitters of the criteria air pollutants that were located in non-attainment counties generally had higher air pollution abatement expenditures, with estimates that imply hundreds of thousands of dollars of additional annual costs for the average affected plant.

Finally, the present study is also related to the literature on the pollution haven hypothesis. This theory suggests that environmental regulation can drive capital in pollution-intensive industries to flee to less regulated locales. This hypothesis was born out of the trade literature, and empirical studies have traditionally focused on cross-country differences in regulation and found weak, if any evidence (for example, see Eskeland and Harrison 2003, or the review in Jaffe et al. 1995). However, such cross-country studies inevitably introduce wide cross-country variations in factors that may affect environmental quality or

regulation, complicating identification. A few recent exceptions (e.g., Fredriksson et al. 2003, Keller and Levinson 2002) have focused on variation in environmental regulation within the United States, thus avoiding the difficulties of comparing different countries, and have found some support for the pollution haven hypothesis.

My study furthers the literature on these topics in three key complementary ways. First, it contributes to the TRI literature by being the first to examine its effects on manufacturing outcomes, specifically new plant births. Furthermore, it advances our understanding of the effects of TRI by overcoming some of the major shortcomings of past studies; it uses data both before and after the introduction of TRI, it captures the effects of all channels of influence, and it focuses on a broader set of industries and firms. Second, it allows for direct comparisons of this relatively new regulatory approach with previous papers that have established the effects of more traditional environmental regulations. Finally, my results shed further light on the question of changes in capital flows in response to tightening of environmental regulations, though at a more micro level – the relevant borders here are between counties, rather than countries.

4 Data

In my empirical analysis I employ confidential establishment-level micro-data accessed via the Michigan Research Data Center of the Census Bureau's Center for Economic Studies (CES). The primary data sets in use will be the Longitudinal Business Database and the Census of Manufactures. I discuss each of these in turn.

4.1 The Longitudinal Business Database

The primary data set I employ in this paper is the Longitudinal Business Database (LBD), built and maintained by researchers at the Center for Economic Studies of the U.S. Census

Bureau. The LBD provides longitudinally linked data for all employer (i.e., those with paid employees) establishments contained in the Census Bureau's business register, the Standard Statistical Establishment List (SSEL), with data dating back to 1975. This allows researchers to observe all establishment births (and deaths) beginning in 1976, as well as data on employment, location, industrial activity, and firm affiliation.

The LBD represents a significant improvement over previous longitudinal datasets compiled by the Census Bureau, which suffered from broken longitudinal linkages that led to spurious establishment births and deaths (Jarmin and Miranda, 2002). Previous studies (Becker and Henderson, 2000; Levinson, 1996) using Census Bureau data to look at the effects of environmental regulation on plant births have focused on the Census of Manufactures (CMF), which provides data on the entire population of U.S. manufacturing establishments every 5 years. List et al. (2003) have pointed out that this 5-year window will fail to capture plant births that are born and die within that span of time. They estimate that this may cause researchers to miss as much as 25% of new manufacturing plant births.

4.2 The Census of Manufactures

I complement my baseline regressions using data from the Census of Manufactures, despite this issue with the data. While the CMF is conducted with lower frequency, the data contained therein is more comprehensive in that it also provides wage and output data that is not contained in the LBD. In addition to using this dataset to check the robustness of my results, I also use CMF data (along with data from its annual sub-sample the Annual Survey of Manufactures) to construct proxies for real wage rates in the counties being analyzed.

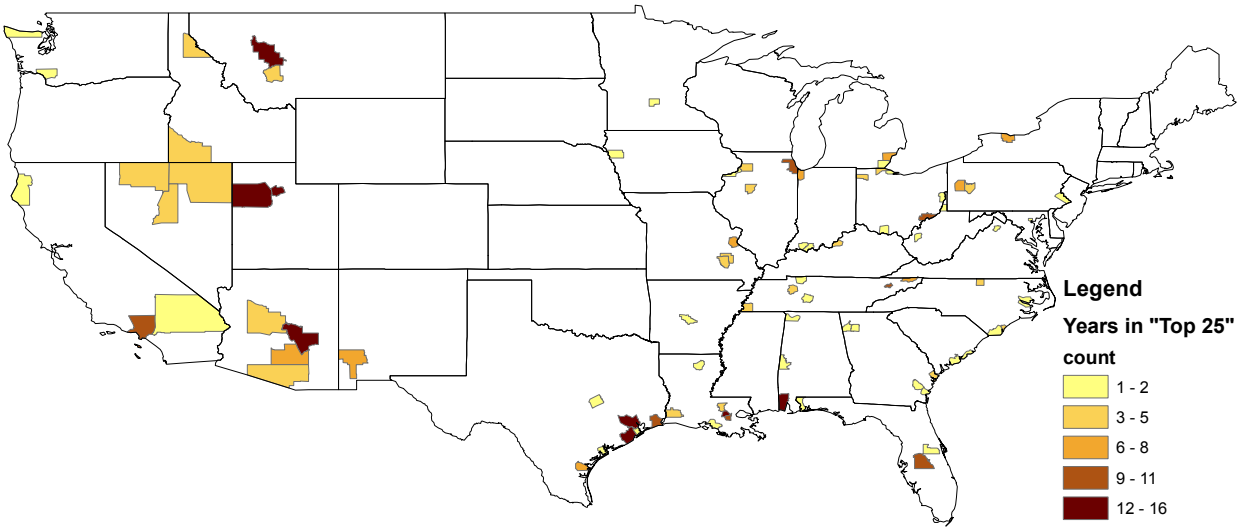


Figure 2: Geographic Distribution of 'Top 25 TRI' treatment

4.3 Supplementary Data Sets

I use Toxics Release Inventory Data to construct a county-level treatment variable. Using TRI data, I replicate the "Top 25" county lists by ranking counties in terms of their un-weighted[9] total of on-site land, water, and air releases of all chemicals. I use these sums to construct an indicator variable that equals 1 if counties are among the "top 25" in on-site releases. There are thus 400 county-year observations (25 counties x 16 years); their geographic distribution is given in Figure 2. This figure suggests that there is significant movement in and out of the "top 25" over time, as only a handful of counties appear in the list 12 times or more. The figure also indicates that there is a significant amount of geographic heterogeneity among the treated counties; there are several treated counties in most regions of the country.

[9]Toffel and Marshall (2004) suggest that when using TRI data as a measure of organizations' environmental performance, using one of several databases to weight the TRI chemicals by toxicity is more appropriate than summing un-weighted releases. However, in my analysis, I argue that the un-weighted sums are most appropriate, as this is how the data were presented in EPA reports. As a robustness check, I estimate alternative specifications where the treatment variable is based on sums of releases weighted by the inverse of the "reportable quantities" established under the Resource Conservation and Recovery Act, and find (unreported) results similar to those presented in later sections of the paper.

I also experiment with using an indicator variable for counties that were in the top 50 and top 100 for on-site releases, as well as a continuous 0 to 1 measure where the county with the highest releases in each year is assigned a value of 1 and the county-level sums are scaled accordingly.[10] Note that the Toxics Release Inventory is generally made available to the public in the spring 2 years after the relevant year; for example, 1987 data first became available in June 1989. I thus allow my treatment measure based on year t data to affect births in year $t + 3$.[11]

I also use TRI data to construct sector-level measures of toxic emissions per employee, where employment data are obtained from the LBD. I use designations from the EPA Sector Notebook Project to define sectors with similar pollution profiles.[12] I use historical nonattainment area designation data from the EPA Green Book in order to control for traditional regulatory pressure as instituted by the 1977 CAAA. Finally, I employ data from the Census Bureau's County and City Data Book series for county-level demographic data.

I use the EPA Sector Notebook Series to classify manufacturing births into one of 21 sectors. I classify these 21 sectors into two groups on the basis of average TRI emissions per employee over the period 1988 to 2005,[13] which are summarized in Table 1. The 8 dirtiest sectors (non-ferrous metals, agricultural chemicals, inorganic chemicals, organic chemicals, plastic resins, oil, pulp and paper, and steel) are grouped in the High-TRI (or "dirty") group, while the next 13 sectors are grouped together in a Middle-TRI group.[14] I also follow Becker

[10]Scaling these measures is necessary because reporting requirements were not constant over time. Regardless of the measure used, all yield qualitatively similar results.

[11]I have experimented with allowing the lag to only be two periods instead of 3. The results are weaker, but generally consistent with those presented here.

[12]The SIC and NAICS designations, which are commonly used as the boundaries defining an industry or sector, are typically based on product definitions. The EPA Sector Notebooks combine 4-digit SIC codes whose industrial facilities and environmental issues are similar.

[13]I omit 1987 for reasons described above, though the results are insensitive to its exclusion. I also check for movement in the relative rankings of the industries over the years, and there is very little movement - the 8 sectors I classify as dirty are the 8 dirtiest, by TRI releases per employee, in nearly every year since the TRI was created.

[14]It could also be argued that there are natural breakpoints between the 4th and 5th sectors, or between the 7th and 8th sectors. I re-constitute the High-TRI group with 4 sectors and 7 sectors and report these

and Henderson (2000) in defining a Control (or "clean") group, comprised of several 4-digit SIC industries which are both (1) non-polluting manufacturing industries that are generally regarded as below the radar of the EPA and (2) not major suppliers to polluting industries. These include all apparel industries (all of SIC code 23), mattresses and bedsprings (SIC 2515), and leather gloves, luggage, and handbags (SIC 315-317).

Table 1: Sector-level averages of TRI releases per employee

Sector Name	**SIC codes**	**TRI releases per employee**
Non-ferrous metals	333-334	5072.1
Agricultural Chemicals	2873-2879	4361.9
Inorganic Chemicals	2812-2819	3154.9
Organic Chemicals	2861-2869	2648.1
Plastic Resins	2821, 2823-24	1443.0
Petroleum Refineries	2911	1273.9
Pulp and Paper	2611, 2621, 2631	1150.5
Iron and Steel	331	472.8
Pharmaceuticals	2833-34	192.2
Rubber	30	127.0
Metal casting	332, 336	99.5
Automobile manufacturing	371	98.4
Wood Furniture	2511-12, 2517-19, 2521, 2531-2541	89.8
Stone, Clay, and Glass	32	73.0
Fabricated metals	34	56.2
Lumber	24	44.6
Aerospace manufacturing	3721-3728	42.4
Textiles	22	35.3
Shipbuilding and repair	3731	29.5
Electronics	36	27.6
Printing	2711-2782	21.4
Control	23, 2515, 315-317	1.8

Source: author's calculations from TRI and LBD data. Releases are in pounds per employee. Sector definitions are from the EPA's Sector Notebook Series. Based on this data, I define high-TRI, or dirty, sectors as the first 8 in this table.

5 Empirical Approach

I perform empirical analysis to estimate the determinants of the pattern of plant "births" across geographic units. As described above, both the LBD and the CMF allow researchers to

results in the appropriate section.

observe how many manufacturing facilities are opened in each industry in a given geographic location over time. I am interested in estimating a model of the general form

$$B_{jt} = B(X_{jt}, f_j + e_{jt}) \tag{1}$$

where B_{jt} is the number of births in county j in time t, $B(\cdot)$ is a generic function, X_{jt} is a vector of county characteristics including regulatory variables, f_j is a county fixed effect, and e_{jt} is an i.i.d. error term.

I follow Becker and Henderson (2000) in estimating the conditional poisson model developed by Hausman et al. (1984). The basic poisson model has a single parameter, λ, which enters the argument for the probability of observing B_{jt} births in county j at time t as follows:

$$prob(B_{jt}) = \frac{e^{-\lambda_{jt}} \lambda_{jt}^{B_{jt}}}{B_{jt}!}. \tag{2}$$

The conditional poisson model allows λ to vary as a function of X_{jt} and f_j:

$$\lambda_{jt} = exp(X_{jt}\alpha + f_j) \tag{3}$$

where α is the parameter vector of interest.

Combining equations 2 and 3 and taking the product over $t = 1, ..., T$, the probability of a sequence of births in a county over time is thus

$$prob(B_{j1}, ..., B_{jt}) = \frac{exp\left(-\sum_{t=1}^{T} \lambda_{jt}\right) \prod_{t=1}^{T} \lambda_{jt}^{B_{jt}}}{\prod_{t=1}^{T} (B_{jt}!)}, \tag{4}$$

while the total births for that county over time can be expressed as

$$prob\left(\sum_{t=1}^{T} B_{jt}\right) = \frac{exp\left(-\sum_{t=1}^{T}\lambda_{jt}\right)\left(\sum_{t=1}^{T}\lambda_{jt}\right)^{\Sigma_{t=1}^{T}B_{jt}}}{\left(\sum_{t=1}^{T} B_{jt}\right)!}. \tag{5}$$

Rearranging terms, the probability of observing a particular sequence of births over time within a county, conditional on the total number of births observed in that county over T periods, can be expressed as

$$prob\left(B_{j1},...,B_{jt}\middle|\sum_{t=1}^{T} B_{jt}\right) = \prod_{t=1}^{T}\left[\frac{exp(X_{jt}\alpha)}{\sum_{s=1}^{T} exp(X_{js}\alpha)}\right]^{B_{jt}} \cdot \frac{\left(\sum_{t=1}^{T} B_{jt}\right)!}{\prod_{t=1}^{T}(B_{jt}!)}. \tag{6}$$

The fixed effect is thus conditioned out of the likelihood, and the estimates are robust to any time-invariant unobserved heterogeneity. The estimation method is thus inherently robust to a primary source of omitted variable bias, and not needing to identify a large number of county fixed effects allows for fairly quick convergence. Counties that have no births over the entire length of the panel in the sector(s) being analyzed are dropped from the sample, since they contribute no information to the likelihood. Following Wooldridge (1999), I calculate standard errors that are robust to violations of the poisson assumption that the conditional mean equals the conditional variance.[15]

After constructing the samples discussed in the previous section, I estimate the conditional poisson model described above. For each specification, I estimate the model twice; once for the group of High-TRI sectors and once for the control sector (which forms its own group, of 1 sector). The X vector includes several controls, which I summarize here.

[15]In unreported results, I also re-estimate my baseline specifications using conditional negative binomial models and obtain quantitatively similar results. For the conditional poisson models, I have also estimated bootstrapped standard errors, which are generally very similar to those reported here.

- $MFGEMP$ is the log of manufacturing employment within the county but *outside* the group, lagged one year (I construct this from the LBD). While the estimation method conditions out all time-invariant observables, it is reasonable to expect that some of the unobservables will change over a 29-year-panel. This variable is meant to capture the effect of any time-varying unobservables (such as market growth, infrastructure improvements, or tax law changes) that would make the county more or less attractive to *all* manufacturing;

- $REALWAGE$ is the log of the real wage rate within the county but *outside* the group, lagged one year. I deflate wages using the output price index for each 4-digit SIC, using the NBER-CES Manufacturing Industry Database by Bartelsman, Becker, and Gray (available at http://www.nber.org). I then take a weighted average within each county of the prevailing real wage rate outside the group being analyzed, and take the natural log.

- NA_{co}, NA_{o3}, NA_{so2}, and NA_{tsp} are indicator variables for non-attainment status for carbon monoxide, ozone, sulfur dioxide, and total suspended particulate matter, respectively. Each equals 1 if the county (in part or in whole) was in non-attainment for the respective pollutant as of January 1st of that year, and 0 otherwise.[16] This allows me to control for any changes in traditional regulatory status that might have affected plant births in polluting industries.[17]

- $OWNEMP$, which I also construct from the LBD, is the log of manufacturing em-

[16]Non-attainment status is actually decided on July 1st of every year, but I assume that this has no effect on plant births until the following year. I have experimented without this 'lag', and the results, while weaker, are consistent with those presented here.

[17]There are two EPA air quality standards that I do not implicitly control for; lead and nitrogen dioxide. First, only 12 counties have ever exceeded the lead quality standards. Secondly, only four counties have ever exceeded NO_2 standards, and any county-year where the NO_2 standards are not met, the ozone standards are not met either. There are also two standards (total suspended particulate matter, in 1991 and ozone, in 2004) whose definition changed slightly during the period under consideration. In both cases, I assume that non-attainment status had the same effect under the new definition as it did under the old.

ployment inside the county and *inside* the group, lagged one year. Plant births will generally not be independent over time, and may be influenced by recent growth or expansion in the same sector - we may see agglomeration effects that encourage entry, or we may see local competitive effects that discourage entry. High levels of employment (or analogously, a high number of facilities) in the group of dirty sectors may also dissuade future entry even in the absence of disclosure, through a "smokestack effect", as stakeholders will have some noisy signal of the level of toxic pollution even in the absence of disclosure. This variable is meant to control for these effects.

- Every regression also includes year dummies, in order to capture any possible unobserved changes that affect all counties in the estimation, such as macro-economic shocks or changes in trade policy.

- Several county-level demographic variables are also included in some specifications, all from the City and County Data Book Series. $INCOME$ is per-capita income in thousands of year 2000 dollars, $UNEMP$ is the prevailing unemployment rate (in percentage terms), $HIGHSCHOOL$ is the percentage of the county's population over 25 years of age that has a high school diploma, $POVERTY$ is the percentage of the county's families below the poverty level, $WHITE$ is the percentage of the county's population that is white, and $TURNOUT$ is the percentage of residents who are of voting age that voted in the most recent presidential election.

Because some of the explanatory variables are based on LBD data (which begins in 1976) and are lagged one year, my panel begins in 1977. It ends in 2005, the most recent year of available LBD data. I lose a handful of observations because of missing county characteristics or wage data, though the vast majority of counties have a full panel of 29 observations. The number of counties included in each estimation varies based on the sectors defining that sample.

The Toxics Release Inventory differs from previous forms of environmental regulation, and these differences provide several reasons to suspect that we may not see the same shifts in plant births. First and foremost, the designation of counties as attainment or non-attainment under the 1977 Amendments was a bright line treatment - new plants were subject to much stricter controls, in the form of minimum technology standards, in non-attainment counties. Consequently the cost of entry rose much faster in those counties than in others. However, under the TRI, there was no such clear distinction. The county-level designation arising from the TRI is instead a listing of the counties with the highest toxic releases.[18] Second, even in those counties that were highlighted as the 'dirtiest', local and state regulators were given neither a mandate to improve this status nor an explicit incentive to do so (such as the withholding of federal highway funds, as is the case under the standards established by the 1977 Amendments to the Clean Air Act). Third, counties are a fairly blunt measure of the area that might be affected by toxic releases; most of the anecdotal evidence of TRI data being used by stakeholders to generate pressure suggests that the spotlight is typically focused on companies or facilities, rather than all the potential polluters in a county.[19] In fact, Oberholzer-Gee and Mitsunari (2006) suggest that economic effects of the TRI may be limited to geographic levels much finer than the county. In short, the change in incentives brought about by the TRI will depend not just on the county borders, but also on the preferences and reactions of the stakeholders at more micro-levels within the county - how likely they are to learn from the newly-disclosed information, and how likely they are to act on it, conditional on having learned of it.

At the same time, the EPA's regular inclusion of lists of the top U.S. counties for TRI total releases in its annual public data releases does suggest that, on average, different counties face differing levels of pressure. These lists could conceivably affect the decisions of local

[18] See, for example, U.S. Environmental Protection Agency (1989), or subsequent public data releases.
[19] This is exacerbated by the fact that counties vary in size significantly.

regulators, in terms of permitting for new entrants. In addition, any public pressure resulting from the TRI would be expected to be higher, on average, in counties that appear on this list. Furthermore, this method of developing the treatment variable has other advantages; it is straightforward to calculate and is analogous to more traditional regulation and would thus provide estimates that are directly comparable to those in other studies. At the same time, drilling down to finer levels of geographic detail is not feasible, as the estimation method I choose requires aggregation at some level, and very few zip codes or even cities will have witnessed multiple births within my sectors, or even sector groupings, over time.

The limitations of using the county-level treatment variable should bias my estimates of the county-level effects of TRI towards 0. Consequently, using the county-level treatment measure is a fairly conservative approach.

As described in the literature review above, several studies have suggested that response to pollution disclosure programs, and environmental outcomes more broadly, vary with community characteristics, such as income levels, education levels, and voter turnout. Thus, in some specifications I allow the treatment to vary with the prevailing levels of certain community characteristics that proxy for a community's ability to leverage the new information into pressure on the plant. In any specification where I include one or more community characteristics interacted with the treatment variable, I also include the community characteristic by itself to capture its baseline effect. However, the community characteristics tend to move very slowly, implying that most of the differences in plant births due to cross-sectional variation in community characteristics will be captured by the (unidentified) fixed effect - the coefficients on the baseline community characteristics will thus capture the effect of changes in these characteristics within a county.

The data set consists of panel data spanning the period between 1977 and 2005. Summary statistics of the data I employ are provided in Table 2, while correlations between the variables used in the analysis are presented in Table 3.

Table 2: Summary Statistics

	Dirty sector sample		*Control sector sample*	
Variable	Mean	Std. Dev.	Mean	Std. Dev.
NA_{co}	0.055	0.229	0.046	0.209
NA_{o3}	0.173	0.379	0.150	0.357
NA_{so2}	0.023	0.150	0.020	0.139
NA_{tsp}	0.089	0.285	0.076	0.265
$MFGEMP$	7.927	1.483	7.657	1.580
$OWNEMP$	3.307	2.867	3.785	2.570
$REALWAGE$	2.221	0.371	2.179	0.380
$UNEMP$	6.797	3.490	6.942	3.597
$INCOME$	13.771	4.198	13.437	4.222
$HIGHSCHOOL$	66.451	13.995	65.202	14.662
$POVERTY$	11.438	6.115	12.203	6.695
$WHITE$	86.739	14.710	86.431	15.485
$TURNOUT$	55.854	9.875	55.931	9.935
observations	53906		65599	
births	13535		89148	
total counties	1927		2382	
years	1977-2005		1977-2005	

Refer to text for variable definitions. The dirty sector sample is comprised of counties that had at least one new plant in any of the 8 high-TRI sectors between 1977 and 2005.

Table 3: Correlations Between Variables Used in the Analysis

	dirty births	NA_{co}	NA_{o3}	NA_{so2}	NA_{tsp}	ln(mfgemp)	ln(ownemp)	ln(rwagerate)	unemp	income	highschool	poverty	white	turnout
dirty births	1													
NA_{co}	0.2364	1												
NA_{o3}	0.2323	0.3649	1											
NA_{so2}	0.0581	0.1465	0.0763	1										
NA_{tsp}	0.1966	0.405	0.321	0.2699	1									
$MFGEMP$	0.3902	0.3636	0.4313	0.09	0.2851	1								
$OWNEMP$	0.264	0.1621	0.2557	0.1526	0.2409	0.3624	1							
$REALWAGE$	0.1574	0.1243	0.2024	0.0579	0.1143	0.3419	0.1673	1						
$UNEMP$	-0.059	-0.0134	0.0298	0.0468	0.1629	-0.1412	0.0669	-0.1459	1					
$INCOME$	0.1953	0.1744	0.2824	0.0054	0.0019	0.3498	0.087	0.5179	-0.3511	1				
$HIGHSCHOOL$	0.1068	0.1547	0.1816	0.0421	0.0536	0.2063	0.0423	0.495	-0.3057	0.7895	1			
$POVERTY$	-0.0832	-0.133	-0.2705	-0.0727	-0.1142	-0.334	-0.0659	-0.3854	0.2846	-0.5802	-0.6202	1		
$WHITE$	-0.1116	-0.0578	0.004	0.0612	0.0471	-0.061	-0.0988	0.0663	-0.1153	0.0607	0.2253	-0.5819	1	
$TURNOUT$	-0.0447	0.0326	0.02	0.0517	-0.0053	-0.1079	-0.0347	0.1455	-0.1252	0.193	0.3513	-0.3115	0.4081	1

Table 4: Plant Birth Means

	Mean	Treatment county mean	Control county mean
New dirty plants	0.251	1.908	0.240
	(0.851)	(3.844)	(0.784)
New clean plants	1.359	25.795	1.235
	(17.925)	(138.890)	(14.912)

In Table 4, I also provide summary statistics on the mean of plant births in the two samples. It suggests that, in both samples, the treatment group and the control group are quite different. The High-TRI counties tend to have higher levels of births in both clean and dirty sectors. However, the empirical analysis helps to identify how much of this variation is due to other factors, whether observed or unobserved, in addition to the treatment. One advantage of the conditional poisson routine is that it estimates relative, rather than absolute, effects of the treatment, so this difference in means should be unimportant as long as the treatment has a similar proportional effect across countries.

6 Results

In this section, I present the results of the conditional poisson estimations as described above.

6.1 Main Results

The baseline results are presented in columns 1 of Tables 5 and 6. First, I focus on Table 5, Column 1, which estimates the effect of being a known high-TRI county on the number of births in the eight dirtiest sectors in the year where the TRI is expected to have an effect. The coefficient on $HighTRI_{25}$ indicates that a 'treated' county could expect a 9 percent reduction in the number of dirty-plant births. However, the coefficient is only significant at a p-value of .15. Recall that conditional poisson estimation conditions out county-level fixed effects and thus effectively removes time-invariant county-level heterogeneity from the estimates, and that I also include (unreported) year fixed effects in all estimations.[20] Thus, the results presented here are akin to those from a differences-in-differences type procedure

[20]Theoretical results have shown that inclusion of indicator-type fixed effects in non-linear estimation routines can lead to bias in the resulting coefficients, in what is called the incidental parameters problem (see e.g., Neyman and Scott 1948. However, Monte Carlo studies (see Heckman 1981) have suggested that the bias is minimal for panel data with 8 or more observations per unit. With as many as 29 observations per unit, any bias in my estimates due to an incidental parameters problem should be minimal.

- the 9 percent reduction is net of any macro-level variation, and reflects changes within treated counties that did not occur in non-treated counties.

Table 5: Conditional Poisson Estimations of Plant Births in the 8 Dirtiest Sectors

	1	2	3	4	5	6
NA_{co}	0.0148	0.0040	0.0158	0.0093	0.0153	0.0088
	[0.0462]	[0.0447]	[0.0463]	[0.0453]	[0.0463]	[0.0459]
NA_{o3}	0.0019	0.0128	0.0003	0.0053	-0.0010	0.0028
	[0.0411]	[0.0412]	[0.0413]	[0.0412]	[0.0410]	[0.0407]
NA_{so2}	0.0682	0.0719	0.0673	0.0693	0.0690	0.0713
	[0.0869]	[0.0857]	[0.0872]	[0.0871]	[0.0866]	[0.0868]
NA_{tsp}	-0.0125	-0.0201	-0.0115	-0.0158	-0.0100	-0.0123
	[0.0449]	[0.0417]	[0.0462]	[0.0453]	[0.0454]	[0.0448]
$MFGEMP$	0.309***	0.306***	0.309***	0.306***	0.304***	0.301***
	[0.0468]	[0.0469]	[0.0468]	[0.0468]	[0.0467]	[0.0466]
$REALWAGE$	-0.0011	0.0055	-0.0018	0.0037	-0.0042	-0.0003
	[0.0385]	[0.0392]	[0.0385]	[0.0392]	[0.0383]	[0.0389]
$OWNEMP$	-0.105***	-0.106***	-0.105***	-0.106***	-0.106***	-0.107***
	[0.0129]	[0.0128]	[0.0129]	[0.0128]	[0.0129]	[0.0129]
$HighTRI_{25}$	-0.0897	1.250***				
	[0.0611]	[0.334]				
$INCOME$		-0.0075		-0.0070		-0.0068
		[0.00792]		[0.00800]		[0.00804]
$HighTRI_{25} * INCOME$		-0.0723***				
		[0.0173]				
$HighTRI_{50}$			-0.0561	0.3430		
			[0.0590]	[0.342]		
$HighTRI_{50} * INCOME$				-0.0214		
				[0.0181]		
$HighTRI_{100}$					-0.0867*	0.3130
					[0.0444]	[0.227]
$HighTRI_{100} * INCOME$						-0.0211*
						[0.0117]
Observations	53906	53906	53906	53906	53906	53906
Number of counties	1927	1927	1927	1927	1927	1927
pseudo R-squared	0.087	0.094	0.085	0.086	0.088	0.090

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors in brackets. Year fixed effects are included in all specifications. Sample includes a total of 13535 births.

Nonetheless, it's possible that the treatment may have coincided with a decrease in general attractiveness of the affected areas to any manufacturing activity. In order to examine this possibility, I run similar estimations using my control sample of clean industries. In these new estimations, the dependent variable is the number of births in any of several industries that have very little pollution (toxic or otherwise) as described above. The baseline results, presented in Column 1 of Table 6, are not consistent with a general downturn in manufacturing in the treated counties. More specifically, a county receiving the $HighTRI_{25}$ treatment could expect a 17% *increase* in the number of clean-plant births, a finding that

Table 6: Conditional Poisson Estimations of Plant Births in the Control Industries

	1	2	3	4	5	6
NA_{co}	0.0971	0.0718	0.0966	0.0696	0.0817	0.0616
	[0.0636]	[0.0533]	[0.0624]	[0.0522]	[0.0579]	[0.0513]
NA_{o3}	0.0552	0.0633	0.0548	0.0636	0.0502	0.0562
	[0.0831]	[0.0868]	[0.0824]	[0.0859]	[0.0792]	[0.0807]
NA_{so2}	-0.0960**	-0.0774*	-0.0900**	-0.0764*	-0.0762*	-0.0727*
	[0.0388]	[0.0411]	[0.0385]	[0.0404]	[0.0394]	[0.0398]
NA_{tsp}	-0.0543	-0.0295	-0.0582	-0.0323	-0.0714	-0.0535
	[0.0801]	[0.0725]	[0.0801]	[0.0727]	[0.0760]	[0.0681]
$MFGEMP$	0.248***	0.200**	0.248***	0.201**	0.255***	0.217**
	[0.0831]	[0.0869]	[0.0827]	[0.0874]	[0.0774]	[0.0868]
$REALWAGE$	-0.0115	-0.00922	-0.0104	-0.00821	-0.00699	-0.00415
	[0.0385]	[0.0375]	[0.0383]	[0.0373]	[0.0382]	[0.0373]
$OWNEMP$	0.107***	0.0991***	0.105***	0.0970***	0.0949***	0.0874***
	[0.0273]	[0.0240]	[0.0260]	[0.0231]	[0.0203]	[0.0197]
$HighTRI_{25}$	0.171***	-1.002***				
	[0.0345]	[0.319]				
$INCOME$		-0.0158*		-0.0152*		-0.0117
		[0.0088]		[0.0086]		[0.0082]
$HighTRI_{25} * INCOME$		0.0588***				
		[0.0157]				
$HighTRI_{50}$			0.180***	-0.682**		
			[0.0262]	[0.271]		
$HighTRI_{50} * INCOME$				0.0420***		
				[0.0128]		
$HighTRI_{100}$					0.235***	-0.671***
					[0.0413]	[0.187]
$HighTRI_{100} * INCOME$						0.0439***
						[0.00810]
Observations	65599	65599	65599	65599	65599	65599
Number of counties	2382	2382	2382	2382	2382	2382
pseudo R-squared	0.639	0.678	0.647	0.677	0.680	0.692

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors in brackets. Year fixed effects are included in all specifications. Sample comprises 89148 births.

is significant at the 1% level. This suggests that counties that were in the top 25 in the country for toxic releases saw a shift in their new manufacturing activity, away from the dirtiest sectors and towards cleaner industries.

Applying these coefficients to the means in Table 4, this suggests that High-TRI counties would have seen 0.18 more dirty plant births per year and 3.77 fewer clean plant births per year in the absence of the treatment.

These baseline results mask a significant amount of heterogeneity in the county-level response to the high-TRI treatment. In column 2 of both Tables 5 and 6, I allow the effects of the High-TRI treatment to vary with county levels of per-capita income. The results are striking. First, I focus on the results in the high-TRI sectors; the coefficient of -.0723 (which is significant at the 1% level) in Table 5 on the interaction between per capita income and the treatment variable suggests that for each $1,000 increase in per capita income of the affected counties, the number of plant births decreases by more than 7%. At the same time, the coefficient of .0588 on the same variable in Table 6, also significant at the 1% level, indicates that per capita income is also an important determinant of the magnitude of this aspect of the response to the TRI treatment.

Readers may notice that the coefficients on $HighTRI_{25}$ have reversed signs in both of the augmented models. However, this is less alarming than it initially appears; the proper interpretation of this coefficient is now the expected effect of the treatment in a county whose per capita income is 0, which is essentially meaningless. But it does suggest that the shift induced by the High-TRI treatment may be *reversed* in poorer counties. Dividing the coefficient on $HighTRI_{25}$ by the coefficient on the interaction term suggests that, in fact, the shift in industrial activity is in the other direction (from clean to dirty) in counties where per capita income levels are below about $17,000.[21] Equally important, this suggests

[21] This turning point is $17,289 in Table 5 and $17,041 in Table 6. All per capita income levels are expressed in year 2000 dollars.

that in richer counties the response to this High-TRI classification is much stronger than the expected effects reported in column 1 of Tables 5 and 6.

This reversal may appear to be a curious result; it is not immediately clear why the revelation of this information would cause more dirty plants and fewer clean plants to open in dirty counties that are also poor. However, if we consider general equilibrium effects, there is an intuitive explanation for this result. Presumably, the counties that earn this dubious honor have some inherent advantages that have led to the arrival of dirty industries in the first place. The knowledge that a county is particularly dirty does not change this. However, richer counties that access and leverage this information become less willing to host (or less attractive to) dirty industry, so new dirty plants are more likely to land in poorer counties that are (1) still attractive to dirty industry, and (2) can not afford to be as 'picky' about new sources of income.

Above, I provide reasons why defining treatment as being among the top 25 counties in toxic releases is the most appropriate approach, but one might still argue that it is a fairly arbitrary measure. Thus, in columns 3-6 of Table 5, I repeat the analysis in columns 1 and 2, except that I use two alternate measures of the treatment: being among the 50 or 100 dirtiest counties in terms of TRI emissions. The results are consistent with those discussed above; on average, there is a modest decrease in dirty plant births, and the magnitude of this decrease depends on per capita income. For both the "Top 50" and "Top 100" definitions of the treatment, the results are weaker than for the "Top 25", and the significance of the results is generally lower, but this is to be expected if the effects are stronger for those counties with the highest emissions levels. Similarly, in columns 3-6 of Table 6, I repeat the analysis for clean births, but using these alternate definitions of the treatment. The results are again very comparable. In separate, unreported regressions I have also used a continuous measure, and I find qualitatively similar results.

The coefficients of the control variables also yield insights. First, in Table 5, note that

none of the coefficients on the non-attainment status variables are significant. This reflects the fact that the sectors that are the highest polluting in terms of toxic releases are not always the highest polluting in the criteria pollutants,[22] and suggests that non-attainment status does not in general deter plant entry in highly toxic industries. The positive and significant coefficient on $MFGEMP$ suggests that plant births in the dirty sectors are responsive to the same time-varying county traits that spur growth in other manufacturing sectors. Wages do not play a consistent role in determining plant births, which is consistent with the fact that most of the sectors classified as dirty here are capital-intensive and perhaps less sensitive to labor conditions than other manufacturing. Finally, the negative and significant coefficient on $OWNEMP$ suggests that there are not aggregation effects in this sector grouping, and in fact that lagged growth in this group of industries decreases plant births in the current period. In Table 6, the results are fairly similar. One exception is that non-attainment status in sulfur dioxide is correlated with a decrease in clean plant births. While never significant, the coefficients on $REALWAGE$ are uniformly negative, which is consistent with the notion that these sectors tend to be more responsive to wages. Finally, I find evidence of aggregation economies in the clean sector, with a positive and highly significant coefficient on $OWNEMP$.

6.2 Robustness Checks

Upon examination of Table 2, some concern arises that the samples used in estimation of the results presented in Tables 5 and 6 are systematically different, and that these differences may be driving the results. To ensure that this is not the case, I re-estimate the baseline conditional poisson models, for both dirty and clean plant births, on a uniform restricted

[22]For example, Greenstone (2002), using the same EPA Sector Notebook definition of sectors, classifies the nonferrous metals sector as a high emitter of carbon monoxide and sulfur dioxide, but not of total suspended particulates or of the antecedents of ozone, while the organic chemicals sector is classified as a high emitter only of the antecedents of ozone. As indicated in Table 1, these are two of the dirtiest sectors in terms of toxic releases.

sample. This sample is limited to counties that had at least one plant birth in each of the three groups - High-TRI, Middle-TRI, and Control. This limits the sample to 49575 observations in 1747 counties. These results are presented in Table 7, and are virtually identical to the results presented in the first two columns of Tables 5 and 6.

Table 7: Conditional Poisson Estimations of Plant Births, Balanced Sample

	dirty births		clean births	
	1	2	3	4
NA_{co}	0.0133	0.00357	0.0978	0.0736
	[0.0462]	[0.0447]	[0.0645]	[0.0544]
NA_{o3}	-0.00115	0.00941	0.0696	0.0767
	[0.0411]	[0.0412]	[0.0837]	[0.0875]
NA_{so2}	0.0679	0.0712	-0.0979**	-0.0803*
	[0.0869]	[0.0857]	[0.0393]	[0.0425]
NA_{tsp}	-0.0121	-0.0191	-0.0521	-0.0276
	[0.0449]	[0.0417]	[0.0806]	[0.0733]
$MFGEMP$	0.315***	0.312***	0.275***	0.228**
	[0.0486]	[0.0486]	[0.0866]	[0.0948]
$REALWAGE$	0.000251	0.0058	-0.00289	4.96E-06
	[0.0392]	[0.0398]	[0.0394]	[0.0382]
$OWNEMP$	-0.0959***	-0.0967***	0.130***	0.121***
	[0.0131]	[0.0130]	[0.0307]	[0.0272]
$INCOME$		-0.00613		-0.0150*
		[0.00796]		[0.00908]
$HighTRI_{25}$	-0.0846	1.335***	0.161***	-0.996***
	[0.0619]	[0.343]	[0.0359]	[0.323]
$HighTRI_{25} * INCOME$		-0.0763***		0.0580***
		[0.0177]		[0.0159]
Observations	49575	49575	49575	49575
Number of counties	1747	1747	1747	1747
pseudo R-Squared	0.090	0.096	0.656	0.691

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors in brackets. Year fixed effects are included in all specifications. Columns 1 and 2 include 13202 (high-TRI) births, while columns 3 and 4 include 86459 (control sector) births.

As mentioned above, I also re-estimate the models using CMF data. These results are possibly biased, due to the fact that the CMF misses between 22% and 24% of plant births, according to the data used here. These results, presented in Table 8, are qualitatively very

similar to what I find using the LBD data. The dirty-sector birth estimations are presented in the first two columns, with the control sector births in columns 3 and 4. The primary difference is that the baseline coefficient on the treatment variable (column 1) is smaller in magnitude and less precisely estimated in the high-TRI sectors, while the baseline coefficient is much higher in the clean sector estimations (column 3). The income interactions are of the same sign and similar magnitude.

Table 8: Conditional Poisson Estimations Using Census of Manufactures Data

	High-TRI sector births		Control sector births	
	1	2	3	4
NA_{co}	-0.128***	-0.138***	0.0771	0.0741
	[0.0496]	[0.046]	[0.0561]	[0.0551]
NA_{o3}	0.0175	0.0325	0.101	0.102
	[0.0443]	[0.0444]	[0.0878]	[0.0873]
NA_{so2}	0.061	0.0481	-0.100***	-0.0853**
	[0.0523]	[0.0531]	[0.0373]	[0.0405]
NA_{tsp}	0.0511	0.0568	-0.0686	-0.0664
	[0.0428]	[0.0393]	[0.0688]	[0.06]
$MFGEMP$	0.231***	0.224***	0.356***	0.356***
	[0.0422]	[0.0419]	[0.0938]	[0.0955]
$OWNEMP$	-0.0856***	-0.0862***	0.0929**	0.0851**
	[0.0131]	[0.0131]	[0.0402]	[0.0362]
$REALWAGE$	0.0562	0.0591	-0.0494***	-0.0475***
	[0.0512]	[0.0516]	[0.0164]	[0.0158]
$INCOME$		-0.00877		-0.00695
		[0.00908]		[0.00979]
$HighTRI_{25}$	-0.0394	0.984**	0.340***	-0.565*
	[0.0829]	[0.473]	[0.054]	[0.312]
$HighTRI_{25} * INCOME$		-0.0610**		0.0465***
		[0.0258]		[0.0162]
observations	10651	10651	14435	14435
counties	1777	1777	2408	2408
pseudo R-squared	0.196	0.220	0.730	0.740

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors in brackets. Year fixed effects are included in all specifications. Columns 1 and 2 include 10121 (high-TRI) births, while columns 3 and 4 include 67981 (control sector) births.

Examination of Table 1 suggests possible alternate cutpoints for the definition of "dirty"

sectors. In particular, the pulp and paper sector has a much higher average emissions per employee ratio than the iron and steel sector, and the gap between the organic chemicals and resins sectors is fairly large. In the first two columns of Table 9, I present the coefficients of interest from conditional poisson estimations where the outcome variable is births in high-TRI sectors, under the respective alternate definitions. The samples have also been adjusted accordingly, omitting counties that saw no births in the relevant sectors over the 29 years of the panel. The results are very similar to the baseline results presented above.

While the baseline results presented in Tables 5 and 6 provide an overview of the effects of this High-TRI treatment on the affected manufacturing sectors, it doesn't tell us anything about how the TRI impacted individual sectors. Thus, in the remaining columns of Table 9, I present the key coefficients from analogous estimations, but where the count of births is sector-specific, and the samples are redefined accordingly. First, note that the general pattern identified in the main results (a decrease in births, where the magnitude of this decrease is correlated with county income levels, in sectors with high levels of toxic pollution) is reflected in the individual sectors. In 5 of the 8 sectors, the expected impact (panel 1) is negative, though only one coefficient is significant. Furthermore, in 7 of the 8 sectors, the negative impact is increasing in income levels, and the coefficient is significant in 4 of the 7 (see panel 2). However, the estimates are generally less precise, making it difficult to make any strong conclusions about impacts on specific sectors. There are exceptions - the impact was fairly pronounced on inorganic chemicals, and to a certain extent, on the nonferrous metals, plastic resins, and iron and steel sectors. However, it is worth noting that the treatment did not have the expected effect on the pulp and paper industry - the signs on both coefficients of interest are reversed, though neither is significant.

I also run two other robustness checks, though I do not report the results here due to space considerations. First, I estimate conditional negative binomial models[23] that follow

[23]The negative binomial distribution is not subject to the Poisson restriction that the conditional mean

Table 9: Alternate samples

	7 dirtiest sectors	4 dirtiest sectors	Nonferrous Metals	Agricultural Chemicals	Inorganic Chemicals	Organic Chemicals	Plastic Resins	Petroleum Refineries	Pulp and Paper	Iron and Steel
					Panel 1: Restricted model					
$HighTRI_{25}$	-0.08	-0.118	-0.25	0.0422	-0.226*	-0.0948	-0.15	0.169	0.247	-0.0403
	[0.0758]	[0.0810]	[0.175]	[0.169]	[0.126]	[0.0998]	[0.153]	[0.211]	[0.171]	[0.0913]
					Panel 2: Expanded Model w/ Income and Interaction					
$HighTRI_{25}$	1.136***	1.418**	0.758	0.75	1.287	1.697*	0.811	2.059**	0.0156	1.597***
	[0.382]	[0.576]	[1.475]	[0.766]	[0.812]	[0.880]	[1.222]	[0.887]	[1.264]	[0.511]
$HighTRI_{25} * INCOME$	-0.0663***	-0.0852***	-0.0559	-0.0382	-0.0879**	-0.0966**	-0.0537	-0.0978**	0.0122	-0.0859***
	[0.0198]	[0.0294]	[0.0755]	[0.0408]	[0.0431]	[0.0446]	[0.0630]	[0.0495]	[0.0631]	[0.0251]
					Panel 3: Sample Statistics					
Observations	47895	41318	14130	23336	20049	18675	14896	10738	13282	30772
Counties	1708	1467	494	821	703	657	518	380	463	1078
Total births	9388	6363	1241	1658	2000	1464	1270	868	887	4147

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors in brackets. Each column presents the coefficients of interest from two estimations, analogous to columns 1 and 2 of Table 6. Year fixed effects are included in all specifications.

the main specification, and the results are qualitatively unchanged. I also interact the Clean Air Act non-attainment status indicators with the treatment of interest to test whether there are 'spillovers' in the sense that the decrease in births in the affected industries and counties following the introduction of the TRI is driven by the same counties being in non-attainment. I find no such evidence, and the coefficients of interest are largely unaffected. The results from these two robustness checks are available upon request.

6.3 Other community characteristics

So far, I have focused on the role of county income levels in explaining the variation in the effects of the TRI on dirty sector births in the dirtiest counties. However, as discussed in the literature review, other studies have found significant impacts of other community characteristics in determining environmental outcomes, both in the absence and presence of environmental regulation. In this subsection, I discuss the results from an expanded analysis intended to investigate the importance of other community characteristics in this particular setting. The estimation results are presented in Table 10.

In column 1, I have repeated the results from column 2 of Table 5, for ease of comparison. In the second column, I present results from a specification that includes the most recent county-level unemployment figure as well as its interaction with my treatment variable. The results suggest that any decrease in dirty plant births in High-TRI counties is mitigated by higher unemployment levels. Column 3 provides the results from an estimation where the specification in column 1 has been augmented by the percentage of the county's population that has received a high school diploma, as well as its interaction with the treatment variable. The results indicate that any decrease in dirty plant births following High-TRI designation

equals the conditional variance. While my use of robust standard errors should address this over-dispersion concern, I estimate conditional negative binomial specifications as a robustness check. However, the conditional negative binomial is prone to its own problems, in that it is not a true fixed-effect method. See Allison and Waterman (2002) for details.

Table 10: Effects of Other Community Characteristics

	-1	-2	-3	-4	-5	-6
NA_{co}	0.00399	0.000914	0.000219	0.00455	-0.0044	-0.00188
	[0.0447]	[0.0448]	[0.0434]	[0.0438]	[0.0430]	[0.0451]
NA_{o3}	0.0128	0.0167	0.0144	0.0306	0.04	0.0163
	[0.0412]	[0.0410]	[0.0411]	[0.0417]	[0.0410]	[0.0407]
NA_{so2}	0.0719	0.0775	0.0686	0.0609	0.0542	0.0729
	[0.0857]	[0.0868]	[0.0829]	[0.0821]	[0.0788]	[0.0857]
NA_{tsp}	-0.0201	-0.0205	-0.0163	-0.0179	-0.00818	-0.0163
	[0.0417]	[0.0418]	[0.0409]	[0.0410]	[0.0412]	[0.0420]
$MFGEMP$	0.306***	0.301***	0.307***	0.272***	0.288***	0.302***
	[0.0469]	[0.0479]	[0.0467]	[0.0467]	[0.0462]	[0.0470]
$REALWAGE$	0.00545	0.00939	0.0199	0.00936	0.0235	0.00706
	[0.0392]	[0.0396]	[0.0407]	[0.0393]	[0.0406]	[0.0392]
$OWNEMP$	-0.106***	-0.106***	-0.108***	-0.111***	-0.111***	-0.106***
	[0.0128]	[0.0128]	[0.0128]	[0.0130]	[0.0128]	[0.0128]
$HighTRI_{25}$	1.250***	0.946**	2.249***	1.081**	1.472**	1.888**
	[0.334]	[0.369]	[0.700]	[0.448]	[0.645]	[0.747]
$INCOME$	-0.00748	-0.00711	-0.00899	-0.0165**	-0.00856	-0.00847
	[0.00792]	[0.00795]	[0.00768]	[0.00812]	[0.00771]	[0.00799]
$HighTRI_{25} * INCOME$	-0.0723***	-0.0694***	-0.0575***	-0.0681***	-0.0712***	-0.0857***
	[0.0173]	[0.0169]	[0.0163]	[0.0167]	[0.0207]	[0.0232]
$UNEMP$		-0.00671				
		[0.00655]				
$HighTRI_{25} * UNEMP$		0.0409*				
		[0.0248]				
$HIGHSCHOOL$			0.00942***			
			[0.00305]			
$HighTRI_{25} * HIGHSCHOOL$			-0.0169**			
			[0.00734]			
$POVERTY$				-0.0226***		
				[0.00667]		
$HighTRI_{25} * POVERTY$				0.00928		
				[0.0204]		
$WHITE$					0.0141***	
					[0.00283]	
$HighTRI_{25} * WHITE$					-0.00317	
					[0.00523]	
$TURNOUT$						0.00319
						[0.00319]
$HighTRI_{25} * TURNOUT$						-0.00768
						[0.00766]
Observations	53906	53906	53906	53906	53906	53906
Number of counties	1927	1927	1927	1927	1927	1927

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors in brackets. Year fixed effects are included in all specifications. Sample includes a total of 13535 births.

is also associated with the education levels of the affected population.

Columns 4 and 5 present results that are designed to address questions of environmental justice. The finding that counties with higher levels of income experience a decrease in dirty-plant births after high pollution levels are revealed, while those with lower levels of income experience an *increase*, is consistent with the classic environmental justice story - that poor and minority populations bear a disproportionate burden of pollution.[24] In column 4, I present the results from a conditional poisson estimation that includes the percentage of the county's population that is below the poverty line and its interaction with the treatment variable, while the results in column 5 come from a specification that includes the percentage of the population that is white, along with the respective interaction. The baseline effects most likely are reflective of the correlations with economic development. The coefficients on the interaction variables, while having signs that are consistent with the environmental justice story, are insignificant.

Finally, in the last column, I present results from a specification that includes voter turnout in the most recent presidential election and its interaction with the treatment indicator. As discussed earlier, some studies (see e.g., Hamilton 1993) have suggested that location of polluting industrial facilities can be affected by the county's propensity for collective action. The resulting coefficient is negative, but not significant.

7 Discussion and Conclusion

Disclosure-based approaches to environmental regulation have proliferated since the creation and apparent success of the Toxics Release Inventory in the late 1980's. In this paper I investigate whether the increased transparency and access to pollution information engendered by the TRI was adequate to induce polluting entities to change their behavior. I find that

[24]See for example Brulle and Pellow (2006).

in counties with high-levels of TRI emissions, the revelation of this information precipitated a modest decrease in plant births in high-TRI industries and an increase in plant births in clean industries. This effect varies significantly across counties, with the effect being much stronger in higher-income counties, and reversed in low-income counties.

The results of the analysis suggest several lessons relevant to understanding the effects of disclosure programs. First, it appears that we should only expect county-wide impacts in those areas that are at the very top of the distribution. The results are generally mitigated as we move past the "top 25" definition of the treatment. This of course does not mean that stakeholders outside of the dirtiest counties are unable to leverage the information in the TRI.

Second, the finding that the response to the treatment is highly variable and largely context-dependent has a variety of interesting implications. From an efficiency standpoint, this may be a very good thing. The debate over environmental federalism[25] has centered on the fact that the marginal costs of pollution and the marginal benefits from the industrial activity that generates that pollution are probably very heterogeneous across (and even within) jurisdictions, and imposition of uniform standards (as is done under the Clean Air Act Amendments) probably leads to inefficiencies. A disclosure program, on the other hand, decentralizes the pollution 'allocation' process, putting more control into the hands of local regulators, stakeholders, and firms. Dirty counties with lower employment or income levels are probably more willing to play host to additional polluting plants than those dirty counties who are already wealthy and near full employment. Of course, this is the other side of the environmental justice coin; some would argue that more pollution in poorer communities reflects injustice, not simply the economist's efficiency gains.

Third, the county-level effects are generally smaller in magnitude than those found in Becker and Henderson (2000). The highest effects I find, for the nonferrous metals and

[25]For example, see Oates (2001).

inorganic chemical sectors, are both smaller and less significant than those they find in all four of the industries they analyze. Of course, the levels of aggregation are different, but this comparison is consistent with the intuition that the information-based approach to environmental regulation is less onerous than the command and control approach embodied in the Clean Air Act Amendments. To the extent that the decreased growth of high-pollution manufacturing in affected counties' activity represent 'costs' of traditional environmental regulation, the comparison of these studies suggests that disclosure is less costly.

Finally, the findings here buttress the growing evidence in support of the pollution haven hypothesis, or at least its domestic analogue. If TRI-driven notoriety effectively tightens environmental regulation, but only in wealthy counties, the changes driving identification in the present study are similar to a tightening of regulations by the developed world while developing countries stand pat. However, unlike the bulk of empirical studies that have searched for support for the pollution haven hypothesis, I am able to examine a discrete measure of capital formation without introducing wide cross-country variations in factors that may affect environmental quality or regulation.

Unfortunately, the analysis performed does not allow me to make any conclusions about the process through which the TRI had its main impacts. Local regulators could be using the information in their permitting decisions, potential entrepreneurs may be responding to an increased threat of public scrutiny in already-dirty locales, or local stakeholders could be leveraging the information in order to influence these two groups. Clearer identification of these channels presents an obvious opportunity for future research – one that could inform our understanding of firm self-regulation, firm location decisions, collective action processes, and the firm-regulator relationship.

This paper has focused only on plant births. While, as previously discussed, this may well be the most likely industrial outcome to be impacted by environmental regulation, the effect of disclosure on outcomes relevant to incumbents, such as investment, changes in the

production process, productivity, and plant closings, is also key to any evaluation of the relative merits of disclosure-based policies, and presents another opportunity for further study.

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